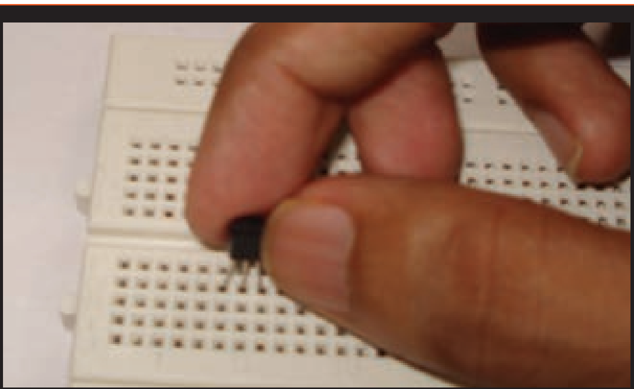
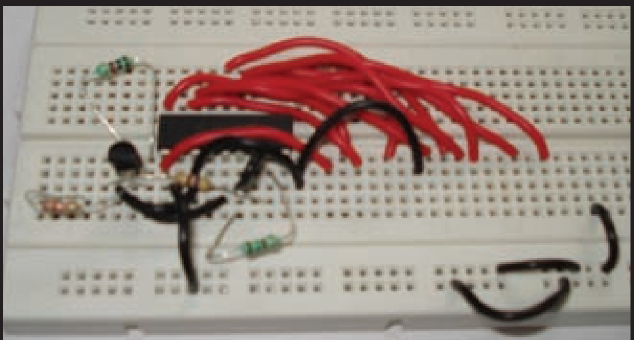
from left to right are: +Vs, Vout and GND.

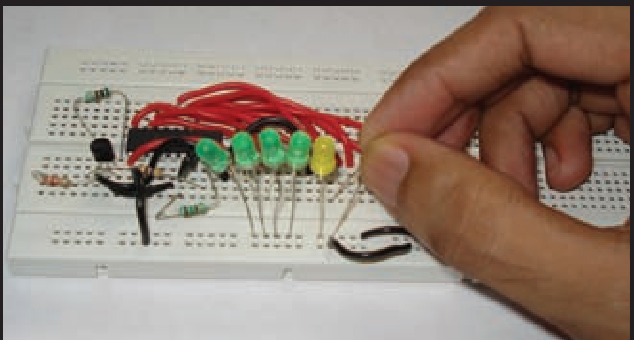
Begin by placing the sensor on the breadboard as shown in the figure. Next place IC LM3914. As you can see from the circuit diagram, you need to connect an array of 10 LEDs to the IC, one to each pin. For this, draw wires from pin nos. 1 and 10 to 18, as shown in the image. The anodes (longer terminal) of the LEDs are all connected together via resistor R8 to positive of battery.

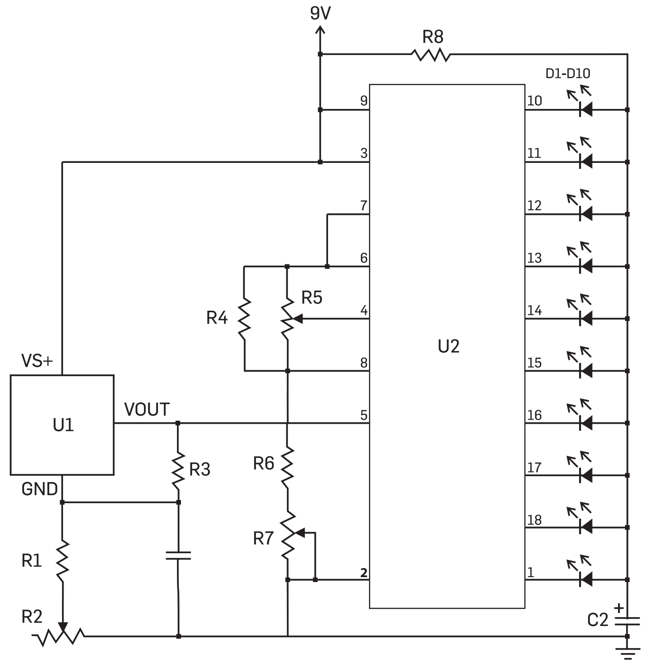
The three potentiometers are connected in three different ways. R2, connected after R1, uses only two terminals (the third terminal is left open). Similarly, R7 also uses only two terminals, but the second and third terminal are connected together. R5 uses all the three terminals as shown in the circuit diagram.

Please ensure that the negative terminals of the electrolytic capacitors are connected to the negative terminal of battery. Reversing the orientation of the capacitors can cause malfunction. This circuit has two modes of operation – bar mode and dot mode. In the bar mode, the LEDs light up resembling a bar graph. Similarly, in the dot mode, the corresponding LED lights up, referring to the specified temperature.


Start by inserting LM35DZ into the breadboard


Draw wires from the pins of LM3914 to connect the LEDs


Now connect the LEDs in series, beginning with green followed by yellow and finally red



Next connect LM 3914 and wires to connect the LEDs

Before you can operate the circuit, you need to calibrate it so that it operates in a linear way. Make all the connections, connect the battery and switch the circuit on. Keep it on for a couple of minutes, so that the circuit stabilises. Measure the voltage at pin 7 of IC2. For this set your multimeter to DCV. Keep the black probe connected to the negative terminal of the battery and the red probe connected to pin 7 of IC2. Vary resistor R7 so that the voltage at pin 6 and 7 stabilises at around 3.345V. Similarly, adjust R5 so that the voltage at pin 4 stabilises at 2.545V.

Now disconnect the battery, remove both ICs from the breadboard and vary R2 such that the resistance across R1 and R2 should be three times the value of R3. For example, if you measure R3 and instead of 1k, it measures 990 ohms, then you need to adjust R2 so that R1 (approx 2.2k) and R2's value would add up to 2.99K.

Reconnect the ICs in the circuit, switch the power on, and the LED should light up as shown in the figure. d

Disclaimer

Although we've taken steps to ensure that the circuit presented here is functional, several factors are beyond our control. In order to avoid malfunction and to avoid dead ends, we recommend assembling the circuit on a breadboard before finally soldering on a circuit board. Procure components as per the list. The project presented here is purely for educational/ entertainment purposes only. We take no responsibility for any loss arising either directly or indirectly as a result of use or implementation of this project.